

Lab 01 — The Price of One Request

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GitHub repository: <https://github.com/Daryn006/lab-01-AI-course>

1. Part 1 Prediction vs. Measured Result

Before running Part 2, I predicted the token ratios for the complaint using UTF-8 byte size.

Language	Prediction	Measured with Gemini 3.6 Flash
Russian / English	1.9×	1.30×
Kazakh / English	2.1×	2.35×

My prediction for Kazakh was close to the measured value, while the Russian prediction was higher than the actual result. This shows that byte length alone cannot exactly predict token count because tokenization also depends on the tokenizer vocabulary and merge rules.

2. Annual Cost

For the production scenario, I assume **2,000 customer-support requests per day**, which represents a medium-sized automated support service.

Language	Input tokens	Billable output tokens	Cost per request	Annual cost
English	101	320	\$0.001276	\$931.30
Russian	131	701	\$0.002727	\$1,990.71
Kazakh	237	944	\$0.003718	\$2,713.96

The total Russian bill was **2.14×** the English bill, while the Kazakh bill was **2.91×** the English bill. At this volume, Kazakh support costs approximately **\$1,782.66 more per year** than English support.

Pricing basis: Gemini 3.6 Flash paid-tier list prices as of September 2026: \$0.75/1M input tokens and \$3.75/1M output tokens.

3. Model for a Kazakh-Language Support Queue

I would use **Gemini 3.6 Flash** for the Kazakh-language support queue. In my test, the model answered fully in Kazakh, did not invent a reason for the interest-rate change when the required documents were missing, clearly stated that it could not determine the reason from the provided information, and suggested contacting official bank support. This gives an acceptable balance between response quality and a measured cost of about **\$0.003718 per Kazakh request**.

4. Additional Cost-Reduction Lever

One cost-reduction method not used in this lab is **prompt caching**, where repeated system instructions can be reused instead of processing the same prompt from scratch for every request.

Core Extension Checks

I also completed the three core extension tasks. My own corpus item showed different RU/EN and KK/EN ratios across the two tokenizers. The Kazakh-specific-letter experiment showed that similar

bytes/char values can still produce very different token counts. In the prose-vs-JSON experiment, JSON added exactly **5 tokens** in English, Russian and Kazakh on both tested tokenizers.

AI Usage Declaration

I used ChatGPT to help me understand the laboratory instructions, troubleshoot Python and Gemini API errors, adapt the measurement scripts for Gemini, and improve the structure and wording of the final report. I executed all laboratory scripts on my own computer and checked the generated token counts and cost calculations myself. ChatGPT was also used to help explain the differences between tokenization, UTF-8 byte size, and final request costs.